Ordinary Amadeus first; enforce the contract afterward

Recording accompanies science. Validation gates acceptance and storage after calculate.

Amadeus scientific calls

download_data → process_covariates → calculate_covariates

Post-calculation export

export_gridmet.R + schema_export.R + storage_boundary.R

Load and verify storage

Typed Parquet readback → DuckDB SQL / fidelity / reopen

Publish and query

Manifest + completion; database.R verifies before read-only open

Record alongside science

provenance_recorder.R observes request / calls / sources / outputs

Build once per release

LinkML / catalog / unit policy → pinned contracts. No runtime AI.

AI assists development / review / documentation only. Humans approve scientific meaning.

Evidence: LATEST

How this differs from the earlier pilot

Fewer workflow gates; shared automatic enforcement at output and storage boundaries.

AREA	EARLIER PILOT	CURRENT STORAGE BOUNDARY
Scientific execution	Pilot orchestration plus workflow checks	Ordinary Amadeus calls; existing checks remain
Integration validation	example_gridmet.R / schema_validator.R path	After calculate: export and shared R / JSON / SQL
Storage acceptance	Version-specific load/check path	Typed readback + load / fidelity / reopen before completion
History and saved queries	Version-specific evidence / protection	Record during science; verify publication at saved open

Validation moved; it did not disappear. Post-calculation work still adds total delivery cost.

Evidence: LATEST, HIST

Close gaps with evidence before broadening scope

Start with policy, rejection coverage and an independent small-request replay.

ORDER	ACTION	REQUIRED EVIDENCE
1	Approve units, time, binding and acceptance	Named / dated policy decisions with exact scope
2	Map critical rules to valid + invalid cases	Rule IDs, source/code hashes and observed outcomes
3	Replay and test publication failure / recovery	Independent identities + values; no false completion
4	Measure cost; qualify deployment and reuse	Matched workload; explicit trust controls and limits

See gap-actions.csv: 11 gap IDs, efficient actions, closure criteria; all pending.